

All practice questions for this topic taken from:

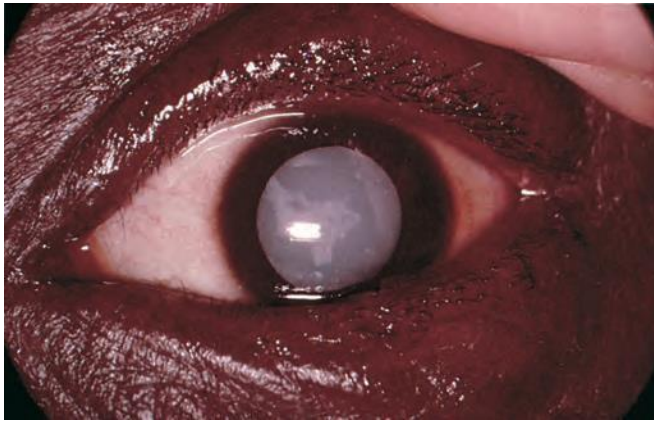
LIPPINCOTT'S ILLUSTRATED Q&A REVIEW OF HISTOLOGY

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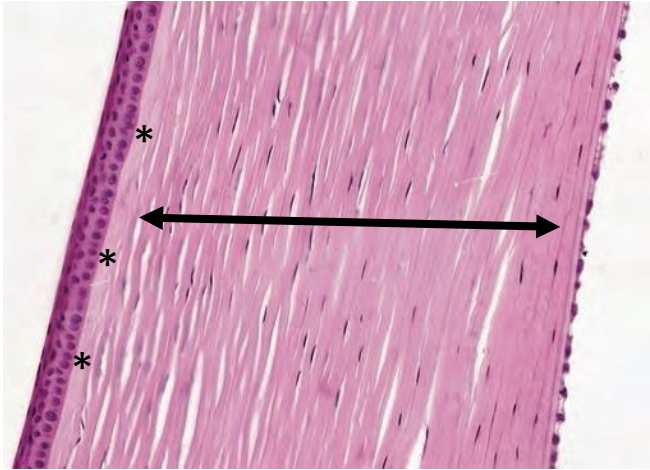
1ST ED.

1. A 56-year-old woman comes to the physician because of a 4-year history of decreased vision in her left eye. Physical examination shows white discoloration of her left as shown in the image below. The white appearance of the pupil in this patient represents a pathologic change affecting a particular structure of the eye. Which of the following components of the eye is most likely affected in this case?



- (A) Conjunctiva
 - (B) Cornea
 - (C) Iris
 - (D) Lens
 - (E) Vitreous body
-
2. You are looking in the mirror and examining your eye. You notice that your pupils are black. Which of the following structures gives a black appearance to the pupils?
- (A) Aqueous humor
 - (B) Iris
 - (C) Lens
 - (D) Uvea
 - (E) Vitreous body

3. A cornea is obtained and examined by light microscopy (shown in the image). Identify the layer indicated by the double arrow.



- (A) Bowman membrane
- (B) Corneal epithelium
- (C) Descemet membrane
- (D) Endothelium
- (E) Stroma

4. A 39-year-old woman afflicted with corneal perforation receives a corneal transplant. Her vision is restored. The corneal graft is well tolerated and becomes fully integrated with the patient's other ocular tissues. Which of the following structural features provides the best explanation for why the corneal graft is not rejected by the recipient?

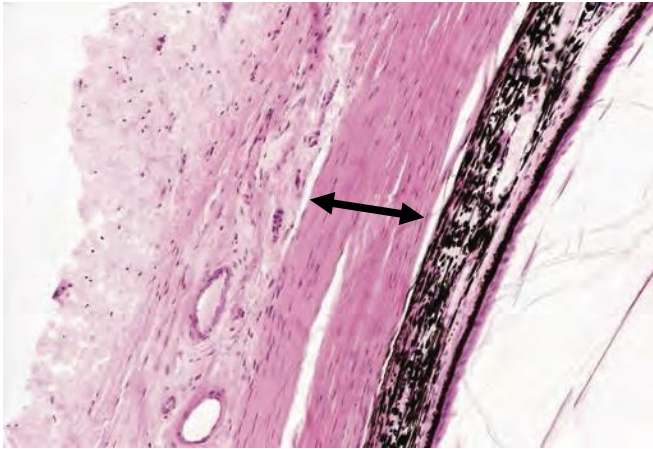
- (A) Basal cells of the corneal epithelium are able to regenerate
- (B) The Bowman membrane protects against the spread of infections
- (C) Collagen fibers are regularly packed
- (D) Cornea is avascular and lacks lymphatic drainage
- (E) Regeneration of the Descemet membrane assists wound healing

5. A 39-year-old woman complains of double vision. During the physical examination, you notice that the patient is unable to abduct her left eye, suggesting paralysis of the lateral rectus muscle. The patient is subsequently diagnosed with abducent nerve palsy. The lateral rectus muscle inserts to which of the following components of the eye?

- (A) Conjunctiva
- (B) Lens

- (C) Retina
- (D) Sclera
- (E) Uvea

6. An eyeball obtained at autopsy is sectioned and stained with H&E. The wall of the globe is shown in the image. Identify the layer indicated by the double arrow.

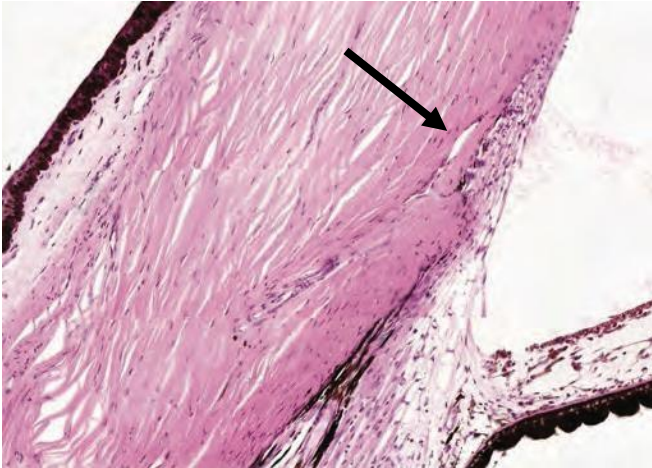


- A. Choroid
- B. Cornea
- C. Retina
- D. Sclera
- E. Uvea

7. A 28-year-old man presents with pain, tearing, and the sense of a foreign object in his left eye. A tiny dust particle is observed and carefully removed. Minor corneal abrasions caused by the dust particle are also noted. After a few days, the scratches on the cornea have completely disappeared. Stem cells that populate the basal layer of the corneal epithelium and facilitate the healing of corneal abrasions originate from which of the following locations?

- (a) Blood vessels in the sclera
- (b) Bowman membrane
- (c) Corneal stroma
- (d) Corneoscleral limbus
- (e) Descemet membrane

8. The corneoscleral limbus of an enucleated globe is examined at high magnification (shown in the image). Identify the channel indicated by the arrowhead.

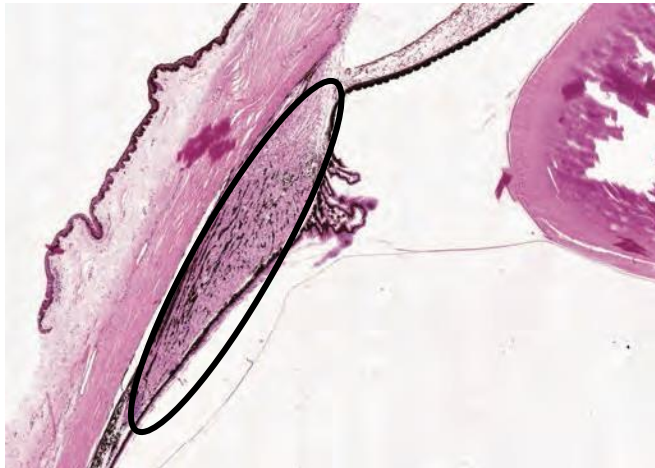


- a. Arteriole
- b. Artifact
- c. Canal of Schlemm
- d. Lymphatic vessel
- e. Trabecular meshwork

9. A 68-year-old woman complains of gradual vision loss over the past year. Ophthalmic examination reveals elevated intraocular pressure. Fundusoscopic examination reveals optic cupping. The patient is subsequently diagnosed with open-angle glaucoma. Which of the following mechanisms of disease is associated with this patient's medical condition?

- A. Corneal abrasion
- B. Degeneration of the choroid
- C. Degeneration of the lens
- D. Increased resistance to the outflow of aqueous humor
- E. Retinal ischemia

10. A section of the anterior segment of the eye obtained at autopsy is prepared with routine H&E staining and examined at low magnification (shown in the image). Identify the structure indicated by the oval.

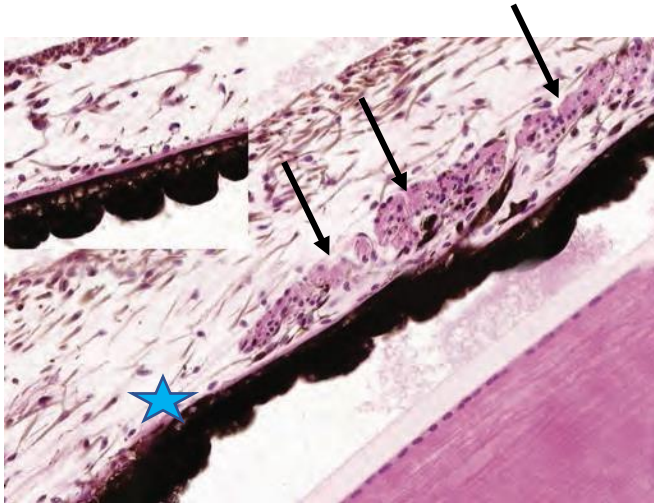


- (A) Limbus
- (B) Ciliary body
- (C) Ciliary process
- (D) Iris
- (E) Zonular fibers

11. A 62-year-old man presents with persistent headaches and gradual visual field loss in his right eye. Ophthalmic examination demonstrates high intraocular pressure. The patient is subsequently diagnosed with closed-angle glaucoma, caused by inadequate drainage of aqueous humor. Which of the following structures in the eye produces aqueous humor?

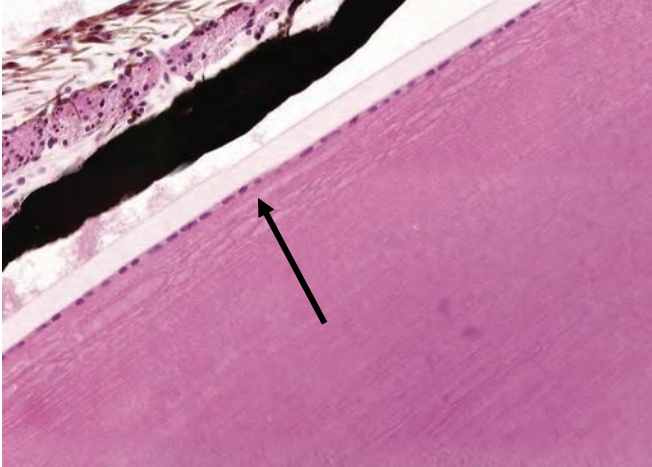
- (A) Choroid
- (B) Corneal endothelial cells
- (C) Corneal epithelial cells
- (D) Non-pigmented epithelial cells of the ciliary process
- (E) Pigment epithelial cells of ciliary process

12. An iris biopsy is examined in the pathology department (shown in the image).
Identify the structure indicated by the arrows.



- (A) Ciliary muscle
- (B) Dilator pupillae
- (C) Iris stroma
- (D) Macrophages
- (E) Sphincter pupillae

13. A group of first-year medical students examine a histologic section of the anterior segment of an eyeball (shown in the image). Identify the cellular layer indicated by the arrow.

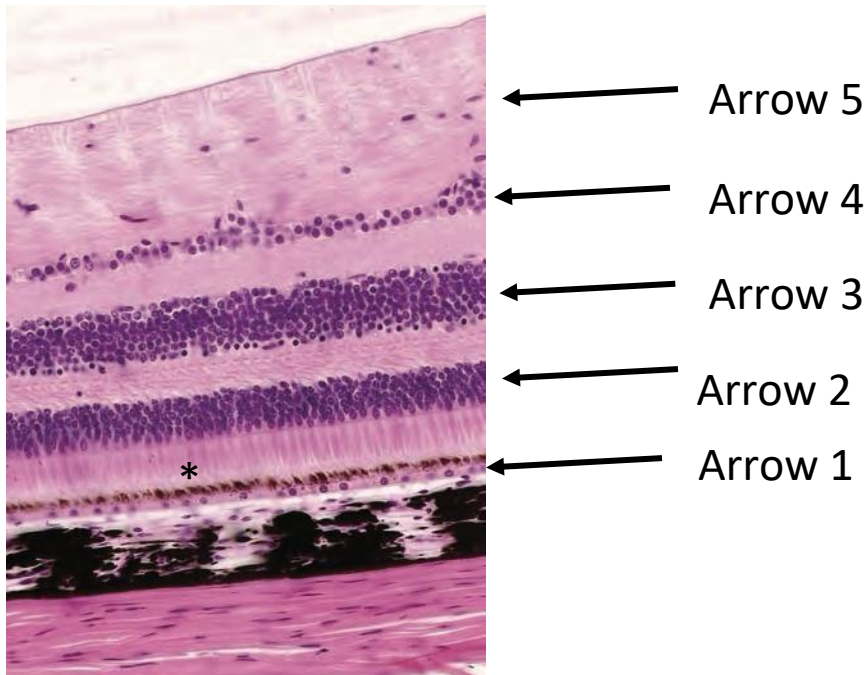


- (A) Developing lens fibers
- (B) Iris epithelium
- (C) Lens capsule
- (D) Lens epithelium
- (E) Mature lens fibers

14. A 48-year-old man loses vision in his right eye after an automobile accident. Fundoscopy reveals intraocular hemorrhage and detachment of the retina. Retinal detachment occurs between which of the following paired tissue layers?

- a. Bipolar cell layer and ganglion cell layer
- b. Bruch membrane and retinal pigmented epithelium
- c. Choroid and Bruch membrane
- d. Outer nuclear layer and inner nuclear layer
- e. Retinal pigmented epithelium and retinal photoreceptor layer

15. During a small group seminar, you are asked to explain how neural impulses originating from an image on the retina are conducted to the brain via the optic nerve. Which of the labeled arrows shown in the image identifies the location of cell bodies of the optic nerve?



- (A) Arrow 1
- (B) Arrow 2
- (C) Arrow 3
- (D) Arrow 4
- (E) Arrow 5

16. A 72-year-old man complains that he has had poor vision for 3 years. Ophthalmic examination reveals a loss of central vision, although the patient's peripheral vision remains unaffected. Which of the following structures is most likely affected?

- a. Cornea
- b. Iris
- c. Lens
- d. Macula lutea
- e. Optic nerve

17. The mother of a 2-year-old boy complains about her son's eye. Physical examination reveals two small red bumps and localized swelling on the outside of the patient's top right eyelid. Acute inflammation of which of the following structures most likely caused the styne observed in this patient?

- (A) Conjunctiva
- (B) Dermis
- (C) Lacrimal gland
- (D) Sebaceous gland of Zeis
- (E) Tarsus

Answers

1. **The answer is D: Lens.** Cataract is a degenerative change of the crystalline lens due to biological aging. The lens is a transparent, biconvex avascular structure suspended in the anterior segment of the eye. It is a refractile component of the eye that helps focus light on the retina. Clouding (opacification) of the lens obstructs the pathway of light to photoreceptors in the back of the eye, leading to vision loss. Cataracts are the major cause of visual impairment and blindness throughout the world. Surgical removal of the cataract lens is the most convenient treatment. Conjunctiva (choice A) refers to the epithelium overlying the sclera and internal surface of the eyelid. Cornea (choice B) is the anterior one-sixth of the external layer of the eye. Iris (choice C) is a contractile diaphragm over the anterior surface of the lens. Vitreous body (choice E) refers to the transparent gelatinous substance in the vitreous chamber that is posterior to the lens. None of these structures are involved in the pathogenesis of cataract. **Keywords:** Cataract
2. **The answer is D: Uvea.** Three concentric layers constitute the wall of the eye (globe). They include (from outer to inner) (1) cornea and sclera, (2) uvea, and (3) retina. There are also three distinct, but interconnected, chambers inside the eye. The lens serves as a boundary for these chambers. The anterior chamber is the space between the cornea and the iris; the posterior chamber is between the iris and the anterior aspect of the lens. The vitreous chamber is between the posterior surface of the lens and the retina. Aqueous humor, a watery fluid, circulates through anterior and posterior chambers. The transparent vitreous body occupies the vitreous chamber. The pupil is the central circular aperture of the iris. When looking at the eye, one looks through the transparent cornea, lens, and vitreous body, toward the heavily pigmented uvea at the back of the eye. The iris (choice B) is usually heavily pigmented, and the color of the iris is referred to as eye color. The other structures are transparent and do not contribute to the black appearance of the pupil. **Keywords:** Eye, pupil
3. **The answer is E: Stroma.** Cornea is the transparent portion of the outer fibrous coat of the eye that covers the anterior one-sixth of the eye. The cornea is about 0.8 to 1.1 mm thick and extremely rich in free nerve endings. The cornea is composed of five layers; from anterior to posterior, they are (1) corneal epithelium, (2) Bowman membrane, (3) stroma, (4) Descemet membrane, and (5) endothelium. Corneal stroma, also referred to as substantia propria, accounts for 90% of the thickness of the cornea and is formed by highly organized collagen fibers. Collagen fibers in the corneal stroma are composed primarily of type I and type V collagen and exhibit a precise, small diameter. The fibers are parallel and arranged into thin layers or lamellae. There are about 60 lamellae in the corneal stroma. The fibers in neighboring lamellae are oriented at approximately right

angles. Slender, flattened fibroblasts are situated between lamellae. The ground substance is rich in proteoglycans containing negatively charged chondroitin and keratin sulfate side chains that serve to covalently bind water. Transparency of the cornea depends on the precise arrangement of collagen lamellae and the affinity for water for extracellular matrix. Nerves reaching the corneal epithelium traverse the stroma, but the stroma itself is avascular.

4. **The answer is D: Cornea is avascular and lacks lymphatic drainage.** Corneal transplantation, also known as penetrating keratoplasty, is the most common and successful tissue transplantation. Cornea is one of the few organs that is not rejected as allografts by the recipient's immune system. Lack of immune rejection contributes to the remarkable success of this surgical procedure. The cornea does not normally contain blood and lymphatic vessels. This prevents the delivery of immunogenic materials to regional lymph nodes and further prevents the migration of activated immune cells to the allograft. In addition, recent studies have shown that major histocompatibility complex (MHC) class II molecules are not expressed in the cornea. The avascular cornea is nourished by aqueous humor in the anterior chamber of the eye. None of the other choices describe a structural basis for the lack of immune rejection of corneal transplants. **Keywords:** Corneal transplantation
5. **The answer is D: Sclera.** The sclera is the tough, opaque part of the outer fibrous coat that covers the posterior five-sixths of the eye. Along with the cornea, the sclera forms the external fibrous skeleton for the eye, providing shape and resistance, as well as attachment sites for extraocular muscles. It is commonly referred to as "the white of the eye," and its anterior part is visible. Diseases of the central nerve system (notably the brainstem) involving the oculomotor and/or abducent nerve may paralyze one or more extraocular muscles. Conjunctiva (choice A) is the transparent mucous membrane covering the internal surface of the eyelids (palpebral conjunctiva) and the anterior part of the sclera (bulbar conjunctiva). Bulbar conjunctiva adheres to the periphery of the cornea. None of the other choices provide attachment sites for the extraocular muscles. **Keywords:** Sclera
6. **The answer is D: Sclera.** Sclera, the thick fibrous coating of the eye, is composed of dense, irregular collagenous connective tissue with bundles of collagen fibers oriented in various directions and in planes parallel to the surface of the eye. The sclera also contains a rich network of elastic fibers. Externally, the sclera is in contact with periorbital adipose tissue. Internally, the sclera faces the choroid (choice A)—a heavily pigmented layer to the right of the indicated sclera. The choroid is a richly vascular sheet within the uvea (choice E). Uvea is the middle layer of the wall of the eye containing blood vessels. A thin layer of loose connective tissue, known as suprachoroid lamina, separates the sclera from choroid. Anteriorly, the sclera is continuous with the cornea (choice B)—the anterior portion of the outer fibrous coat of the eye that exhibits dense regular connective tissue in its stroma. Posteriorly, the sclera is continuous with the dura

mater covering the central nervous system. Retina (choice C) is the inner layer of the eye that contains photoreceptor cells and multiple layers of neurons. Histologic features of the retina are not evident, because the image shows only the anterior portion of the eye. **Keywords:** Sclera

7. **The answer is D: Corneoscleral limbus.** The cornea and sclera are continuous. The transitional zone encircling the cornea is termed the corneoscleral limbus or corneoscleral junction. This zone is where the transparent cornea merges with the opaque sclera. Stem cells for the corneal epithelium are concentrated at the limbus. From this stem cell niche, rapidly dividing cells move centripetally into the corneal epithelium. Minor lesions of cornea can heal rapidly without treatment due to the significant regenerative capacity of the cornea. The stem cells also serve as a barrier to conjunctival epithelial cells to prevent them from migrating into the cornea. Depletion of the stem cell niche at the limbus, due to disease or injury, may result in an abnormal corneal surface and “conjunctivalization” of the cornea. Basal cells of the corneal epithelium are recruited from the stem cells and further differentiated into squamous cells moving toward the surface. Other listed choices do not provide stem cells for the corneal epithelium. **Keywords:** Corneoscleral limbus

8. **The answer is C: Canal of Schlemm.** At the inner aspect of the limbus region, close to the angle between the cornea and the iris, the corneal endothelium and Descemet membrane are replaced with a plexus of irregular endothelium-lined channels, referred to as trabecular meshwork (choice E). Adjacent and deep to these channels, a larger space (the canal of Schlemm or scleral venous sinus) is situated in the stroma of the limbus. The canal of Schlemm forms a complete circle at the apex of the anterior chamber angle. The canal of Schlemm and trabecular meshwork form an apparatus for the outflow of the aqueous humor. Aqueous humor from the anterior chamber is slowly and continuously drained into the trabecular meshwork, then enters the canal of Schlemm, and eventually returns to veins of the sclera. The endothelial lining of the space indicates that it is not an artifact. None of the other choices exhibit histologic features of the canal of Schlemm. **Keywords:** Canal of Schlemm, aqueous humor circulation

9. **The answer is D: Increased resistance to the outflow of the aqueous humor.** Glaucoma refers to a collection of disorders resulting from increased intraocular pressure. Glaucoma features an optic neuropathy accompanied by a characteristic excavation of the optic nerve head (cupping) and a progressive loss of visual field sensitivity. Primary glaucoma may develop with no apparent underlying eye disease and can be subdivided into open-angle and closed-angle glaucoma. In open-angle glaucoma, the anterior chamber angle is open and appears normal, but increased resistance to the outflow of the aqueous humor due to the degeneration or obstruction of the trabecular meshwork is present within the vicinity of the canal of Schlemm. Deficient aqueous humor absorption leads to increased resistance and intraocular pressure. None of the other choices cause glaucoma.

Keywords: Glaucoma

10. **The answer is B: Ciliary body.** Ciliary body is the expansion of the choroid at the level of the lens. It forms a thickened tissue ring lying inside the anterior part of the sclera. The connective tissue stroma of the ciliary body consists of a thick, outer layer of smooth muscle (ciliary muscle) and an inner vascular layer. Ciliary processes (choice C) are radial ridges extending from the vascular layer of the ciliary body. The surfaces of ciliary processes and the ciliary body facing posterior/vitreous chambers are covered by ciliary epithelium—a double layer of low columnar epithelial cells. A circular system of zonular fibers (choice E) extends from the surface epithelial cells in grooves between ciliary processes and attaches to the lens capsule. This structure is also referred to as the suspensory ligament of the lens. Contraction of ciliary muscles decreases the diameter of the ciliary body ring and reduces tension on the zonular fibers. This enables the lens to become rounded and to focus light from nearby objects onto the retina. Continuous contraction of ciliary muscles (e.g., prolonged reading) may result in muscle spasm, loss of visual accommodation, and myopia. None of the other structures exhibit histologic features of the ciliary body. **Keywords:** Ciliary body
11. **The answer is D: Nonpigmented epithelial cells of the ciliary process.** The cells of the nonpigmented layer of the ciliary epithelium demonstrate characteristic features of fluid-transporting cells, namely tight junctions, extensive basal infolding, and Na–K pumps in their lateral plasma membranes. These cells transport fluid from the vascular stroma of the ciliary processes into the posterior chamber, located between the posterior surface of the iris and the anterior surface of the lens. The ionic composition of aqueous humor is similar to plasma, except for reduced protein content (0.1% in aqueous humor vs. 7% in plasma). Aqueous humor in the posterior chamber flows through the pupil into the anterior chamber of the eye, where it nourishes the lens and cornea. Aqueous humor is drained through the trabecular meshwork located at the iridocorneal angle and returned to veins through the canal of Schlemm. In closed-angle glaucoma, the iridocorneal angle is completely closed due to anterior displacement of the iris root against the cornea, thereby obstructing the outflow of aqueous humor through the trabecular meshwork. This leads to an accumulation of fluid and increased intraocular pressure. None of the other structures produce aqueous humor. **Keywords:** Glaucoma, ciliary process, aqueous humor
12. **The answer is E: Sphincter pupillae.** Iris is the most anterior extension of the choroid that forms a contractile diaphragm with a central aperture (pupil). The peripheral portion of the iris partially covers the lens. The uveal portion of the iris is composed of loose connective tissue, with abundant melanocytes and fibroblasts (choice C). The number of pigmented melanocytes determines the color of the iris. Near the margin of the iris, an involuntary sphincter pupillae muscle is arranged in a circular array. The posterior aspect of the iris is lined by a double layer of cuboidal epithelial cells that represent the anterior continuation of

the ciliary epithelium. The cells of both layers are pigmented, and this heavy pigmentation ensures that light enters the eye only through the pupil. The deep layer consists of myoepithelial cells that constitute the dilator pupillae (choice B, inserted image, indicated by the star). Contractions of dilator and sphincter muscles regulate the diameter of the pupil and control the amount of light entering the eye. Ciliary muscle (choice A) is smooth muscle found in the ciliary body.

Keywords: Iris, sphincter pupillae, dilator pupillae

13. **The answer is D: Lens epithelium.** The lens is a transparent, biconvex, elastic, and avascular structure situated immediately behind the iris and pupil. The bulk of the lens consists of elongated concentric lens fibers. The lens epithelium is a monolayer of cuboidal epithelial cells covering the anterior surface of the lens. They are referred to as a subcapsular epithelium, because a thick basal lamina (elastic lens capsule) covers the epithelium. The subcapsular epithelial cells near the equator of the lens proliferate and differentiate into new lens fibers (choice A), a process responsible for the growth of the lens and continuous deposition of new fibers throughout life. As the new lens fibers mature (choice E), they move toward the center of the lens. Lens capsule (choice C) is a thick, homogeneous membrane that provides an attachment site for zonular fibers. Iris epithelium (choice B) is seen on the image as a pigmented epithelium covering the iris.

Keywords: Eye, lens

14. **The answer is E: Retinal pigmented epithelium and retinal photoreceptor layer.** Retina is the innermost layer of the wall of the globe. It is derived from the optic cup and is composed of two layers: (1) the inner neural retina (light sensitive photoreceptors and conducting neurons) and (2) the outer non-sensory retinal pigmented epithelium, which firmly attaches to the choroid through the Bruch membrane. A potential space is present between the pigmented epithelium and the photoreceptor cell layer of the sensory retina. Traumatic hemorrhage can cause separation between these two layers, leading to retinal detachment. This is a medical emergency that requires immediate treatment. Blindness and permanent damage may occur if retinal detachment is not repaired urgently. Retinal defects, diminished pressure on the retina, and vitreous traction can also lead to this medical emergency. None of the other tissue layers are associated with retinal detachment. **Keywords:** Retinal detachment

15. **The answer is D: Arrow 4.** The visual apparatus focuses light on the retina, which converts the image to neural impulses that are sent to the brain via the optic nerve (cranial nerve II). As seen in the image, the retina is a multilayered structure with three major cellular layers: (1) outer nuclear layer (arrow 2) containing nuclei of photoreceptor rod and cone cells; (2) inner nuclear layer (arrow 3) composed of bipolar neuron cell bodies; and (3) an internal ganglion cell layer (arrow 4). Regions containing the fibers and synapses between adjacent inner and outer nuclear layers are referred as outer plexiform layers. The inner

plexiform layer contains bipolar axons and the dendrites of ganglion cells. Ganglion cells are large, multipolar neurons with their cell bodies located in the cellular ganglion cell layer. Their axons form the nerve fiber layer (arrow 5) that is internal to the ganglion cell layer. These nerve fibers converge at the optic disc to form the optic nerve. The retina also contains supporting glial cells, Müller cells, and other association neurons (horizontal neurons and amacrine neurons). Arrow 1 indicates pigmented epithelial cells of the retina. **Keywords:** Retina, optic nerve

16. **The answer is D: Macula lutea.** Macula lutea surrounds the fovea and protects cone cells of the fovea. With advanced age, the retina around the macula lutea may undergo degenerative changes resulting in age-related macular degeneration (ARMD). ARMD is the most common cause of impaired visual acuity and blindness in the elderly population. Degenerative lesions in the area of macula lutea involve focal thickening of the Bruch membrane, as well as atrophy and depigmentation of the retinal pigmented epithelium and obliteration of capillaries in the choroid layer. Eventually, photoreceptor cells are affected and a blind spot is produced. None of the other structures are involved in the pathogenesis of ARMD. **Keywords:** Macular degeneration

17. **The answer is D: Sebaceous gland of Zeis.** Anteriorly, the eye is protected by the eyelids, as well as the conjunctiva and tears produced by the lacrimal gland. The eyelid is lined by skin externally and conjunctiva internally. Muscle fibers of the orbicularis oculi are situated in the dermis. The tarsal plate (tarsus) is a supportive fibroelastic tissue facing the conjunctiva. Tarsal glands (meibomian glands) are long sebaceous glands embedded in the tarsus. Eyelashes are present at the margins of the eyelids. The sebaceous glands of Zeis and sweat glands of Moll are associated with hair follicles of the eyelashes. Infections of sebaceous glands of Zeis or glands of Moll, generally caused by *Staphylococcus aureus*, lead to external styes on the outside of the eyelids. Infections near the openings of the ducts of the tarsal glands of meibomian can cause internal styes that appear as red bumps beneath the eyelid. None of the other structures are involved in localized styes. **Keywords:** Stye, eyelid